

E11 — Baseline Optimizer Comparison##
Experiment Overview This experiment compares the **Muon** optimizer against four standard first-order optimizers on the Matrix Sensing (MS) problem:- **Muon-Exact**: Spectral normalization via full SVD (our method)- **SGD**: Standard Stochastic Gradient Descent with momentum- **Adam**: Adaptive moment estimation- **RMSprop**: Root Mean Square propagation- **Momentum-SGD**: SGD with Polyak momentum The goal is to determine whether Muon's spectral normalization approach provides measurable advantages in convergence speed and solution quality compared to widely-used baselines. **Problem Setting**: Matrix Sensing with $d=50$, $r=5$, learning rate 0.01, noiseless observations, hard-cutoff spectrum, 2000 iterations, 10 random seeds.

Scientific Question **Hypothesis**: Muon's spectral normalization (which replaces gradient directions with their unit-nuclear-norm counterparts via SVD) leads to faster convergence in terms of iterations-to-epsilon (K_{ϵ}) compared to standard first-order methods on low-rank matrix sensing problems. **Null Hypothesis (H_0)**: There is no significant difference in K_{ϵ} between Muon and baseline optimizers. **Key Metrics**:- K_{ϵ} : Iterations to reach $\text{loss} \leq \epsilon$ (primary metric)- min_loss : Best loss achieved

during optimization- time_s: Wall-clock time in seconds- F_eps: Total FLOPs to reach convergence- I_conv: Convergence flag (1=converged, 0=not)

Experimental Design | Parameter | Value | |-----
---|-----| | Problem | Matrix Sensing (MS) | |
Algorithms | Muon-Exact, SGD, Adam, RMSprop, Momentum-SGD | | Dimension d | 50 | | Target rank r | 5 | | Measurements m | $2dr = 500$ | |
Learning rate | 0.01 | | Noise level | 0.0 | | Max iterations | 2000 | | Random seeds | 10 (0-9) | |
Convergence threshold epsilon | 0.01 | The experiment uses a paired design: each algorithm runs on the same 10 problem instances (controlled by random seed), enabling paired statistical comparisons.

Data Loading & Inspection

```
In [ ]: import pandas as pdimport numpy as npimport matplotlib.pyplot as pltimport s
```

Exploratory Data Analysis### Summary Statistics by Algorithm

```
In [ ]: summary = df.groupby('algo').agg(      n=('seed', 'count'),      K_eps_mean=('K_
```

Key Observations from EDA1. **Convergence Rate:** Muon-Exact achieves the lowest mean K_epsilon (~41 iterations), followed closely by Adam (~42-43). SGD and Momentum-SGD require ~47-48 iterations. **RMSprop fails to converge** within 2000 iterations on all seeds (K_epsilon capped at 2001).2. **Solution Quality:** SGD and Momentum-SGD achieve dramatically better minimum losses (~1e-32) versus Muon-Exact and Adam (~1e-3). This suggests Muon converges quickly to a moderate-precision solution but plateaus earlier.3. **Convergence Flag:** All algorithms except RMSprop achieve I_conv=1.4. **FLOPs:** Despite fewer iterations, Muon's per-iteration SVD overhead results in higher total FLOPs than SGD variants.

Comparative Analysis### Ranking by Performance

```
In [ ]: print('='*70)print('ALGORITHM RANKINGS')print('='*70)print('\n1. By K_epsilon')
```

Visualizations

Plot 1: K_epsilon by Algorithm (Grouped Bar Chart)

```
In [ ]: fig, ax = plt.subplots(figsize=(12, 7))algo_order = ['Muon-Exact', 'SGD', 'A
```

Plot 2: Minimum Loss by Algorithm (Log Scale Boxplot)

```
In [ ]: fig, ax = plt.subplots(figsize=(12, 7))min_losses = [df[df['algo'] == a]['mi
```

Plot 3: Wall-clock Time by Algorithm

```
In [ ]: fig, ax = plt.subplots(figsize=(12, 7))time_means = [df[df['algo'] == a]['ti
```

Plot 4: Total FLOPs by Algorithm

```
In [ ]: fig, ax = plt.subplots(figsize=(12, 7))flops_means = [df[df['algo'] == a]['F
```

Statistical Tests### Pairwise t-tests: Muon-Exact vs Each Baseline on K_epsilon
We conduct paired t-tests comparing Muon-Exact against each baseline. Since all algorithms use the same seeds, the paired design controls for problem instance variability.

```
In [ ]: from scipy.stats import ttest_rel, wilcoxondef cohens_d_paired(x, y): dif
```

```
In [ ]: print('='*80)print('WILCOXON SIGNED-RANK TESTS: Muon-Exact vs Baselines (K_e
```

Conclusions & Interpretation### Summary of Findings
1. Convergence Speed (K_epsilon):
Muon-Exact converges in the fewest iterations (K_epsilon ~ 41), marginally faster than Adam (~ 42-43) and clearly faster than SGD variants (~ 47-49). RMSprop **completely fails to converge**

within 2000 iterations.

2. Solution Quality (min_loss): SGD and Momentum-SGD achieve dramatically better final loss values ($\sim 1e-32$ vs $\sim 1e-3$ for Muon and Adam). This suggests that while Muon converges *quickly* to an approximate solution, it plateaus before reaching the global optimum precision that SGD achieves. The spectral normalization may act as implicit regularization.

3. Statistical Significance: Muon's iteration advantage over SGD and Momentum-SGD is statistically significant (paired t-test, $p < 0.01$) with large effect sizes. The advantage over Adam is not statistically significant.

4. Computational Cost: Muon's per-iteration SVD overhead results in higher total FLOPs than SGD. However, wall-clock time differences are modest ($\sim 33s$ for Muon vs $\sim 28s$ for SGD variants).

5. Adam: Adam shows competitive convergence speed but, like Muon, does not achieve the ultra-low loss values of SGD.

Implications- RMSprop is not suitable for this matrix sensing problem class.- Muon provides meaningful convergence acceleration but the $O(d^3)$ SVD cost is a limiting factor.- For ultra-high precision, SGD may be preferred; for fast moderate-precision convergence, Muon or Adam are competitive choices.